Lecture 1 CPS

Descriptive Regression with a Binary Covariate

Outline

- Univariate Descriptive Statistics
- Descriptive Regression with a Binary Covariate

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- Having the whole population is becoming more common in the era of Big Data.
- Note: if you are interested in the causal effects of a treatment (a drug, a policy, etc.), you can never really have the whole population (we will discuss this in a few weeks).

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From the google definition:

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     - I'm using "information" in a loose sense here, will formalize a bit later
     - reducing the data means taking the data, which are often represented by $n * v$ numbers (with n the number of observations and v the number of variables) and reducing to a descriptive statistic that is represented by a few numbers.

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Enough forest. Trees (examples) are coming!

"Hold on, where's the forest again?"

It is easy in a statistics class to focus on the specifics. I want you to remember why we are doing what we are doing.

Outline

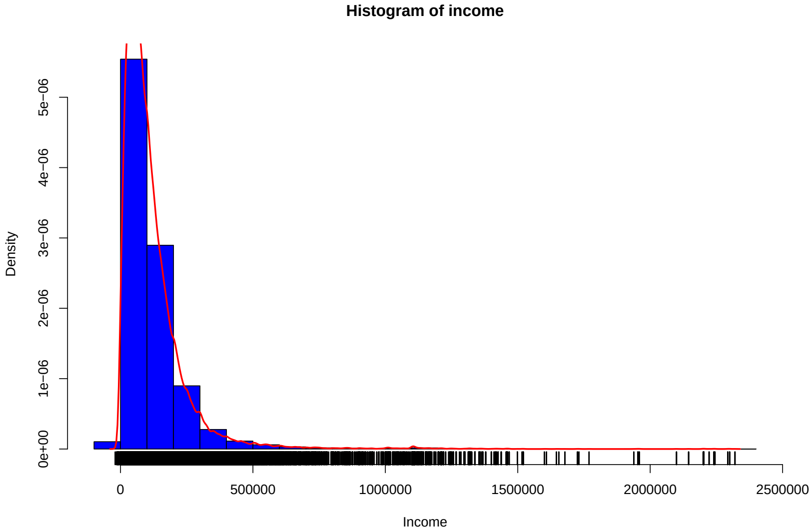
Univariate Descriptive Statistics

Univariate Example: CPS Wage Data

- Data on 2022 household income from the CPS

Household	income
1	44220
2	82752
152732	32300

Describing Univariate Data



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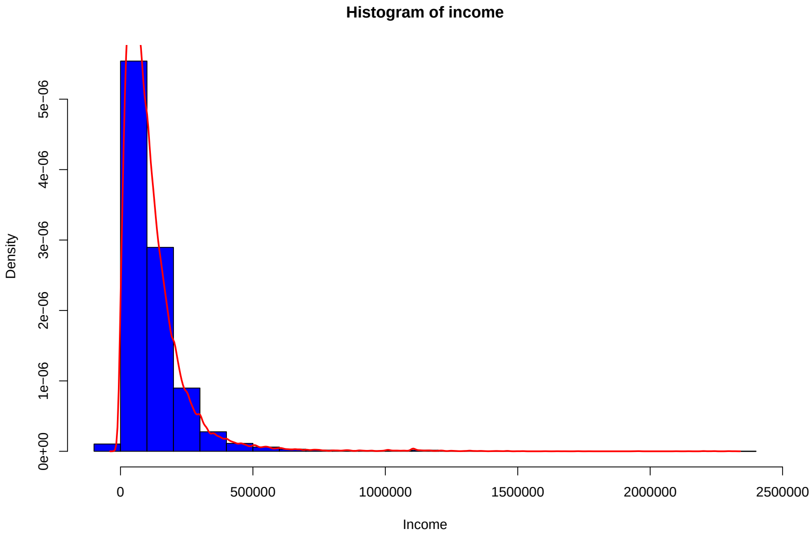
- We typically focus on measures of center and spread.
- We will discuss the rationale for this over the coming weeks.

Univariate Activity



Find the rough location of the mean and median.

Univariate Activity Solution